

Monodromy relations for string amplitudes on AdS space

Luis F. Alday^{1,*}, Rodrigo S. Pitombo^{1,2,†} and Aurélie Strömholm Sangaré^{1,‡}

¹*Mathematical Institute, University of Oxford, Andrew Wiles Building, Radcliffe Observatory Quarter, Woodstock Road, Oxford, OX2 6GG, United Kingdom*

²*ICTP South American Institute for Fundamental Research, Instituto de Física Teórica UNESP Rua Dr. Bento Teobaldo Ferraz 271, 01140-070, São Paulo, SP, Brazil*



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We consider the Anti-de Sitter (AdS) Veneziano amplitude describing the scattering of four gluons in type IIB string theory on $\text{AdS}_5 \times S^3$. We propose a set of powerful monodromy relations between different color-ordered amplitudes. These relations arise as a consequence of the emergent worldsheet description of open string scattering on AdS. In flat space, they reduce to the usual monodromy relations for the Veneziano amplitude, and they hold order by order in the small curvature expansion. The relations hold for the results available in the literature, including the scattering of arbitrary Kaluza-Klein modes.

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Open string scattering on AdS. For holographic Anti-de Sitter (AdS)/conformal field theory (CFT) systems one can define string scattering on AdS through correlators of local operators in the dual CFT. Massless scattering amplitudes correspond to correlators of protected operators and the tree-level approximation corresponds to a large central charge limit [1]. The study of gluon scattering on curved backgrounds through AdS/CFT was initiated in [2]. In the simplest setup one adds probe branes to the standard AdS/CFT system [3–5]. The scattering of open strings attached to these D-branes is dual to the correlator of currents in the CFT. More specifically we focus on the setup of [6,7]. In those papers the AdS Veneziano amplitude for type IIB gluon scattering in $\text{AdS}_5 \times S^3$ was considered, and the first two curvature corrections around flat space were computed. The corresponding CFT is a 4D $\mathcal{N} = 2$ superconformal field theory with $SU(2)_R \times U(1)_R$ R symmetry and $SU(2)_L \times G_F$ flavor symmetry. The AdS Veneziano amplitude maps to the four-point correlator of $\mathcal{O}^I(x, v)$, the superconformal primary of the flavor multiplet. This is a Lorentz scalar operator, with protected dimension $\Delta = 2$ and a singlet of $SU(2)_L$. Furthermore, it transforms in the adjoint of $SU(2)_R$, with polarization vector v^α , and the adjoint of G_F , with index I . The four-point correlator takes the form

$$\langle \mathcal{O}^{I_1}(x_1, v_1) \cdots \mathcal{O}^{I_4}(x_4, v_4) \rangle = \frac{v_{12}^2 v_{34}^2}{x_{12}^4 x_{34}^4} G^{I_1 I_2 I_3 I_4}(z, \bar{z}, \alpha),$$

where we have introduced the cross ratios

$$\frac{x_{12}^2 x_{34}^2}{x_{13}^2 x_{24}^2} = |z|^2, \quad \frac{x_{14}^2 x_{23}^2}{x_{13}^2 x_{24}^2} = |1 - z|^2, \quad \alpha = \frac{v_{13} v_{24}}{v_{12} v_{34}},$$

with $x_{ij} = x_i - x_j$ and $v_{ij} = \epsilon_{\alpha\beta} v_i^\alpha v_j^\beta$. The solution to the superconformal Ward identities can be written in terms of a reduced correlator $H^{I_1 I_2 I_3 I_4}(U, V)$ such that

$$G^{I_1 I_2 I_3 I_4}(z, \bar{z}, \alpha) = G_0^{I_1 I_2 I_3 I_4}(z, \bar{z}, \alpha) + (1 - z\alpha)(1 - \bar{z}\alpha) H^{I_1 I_2 I_3 I_4}(z, \bar{z}),$$

where $G_0^{I_1 I_2 I_3 I_4}(z, \bar{z}, \alpha)$ encodes the contribution from protected operators and $H^{I_1 I_2 I_3 I_4}(z, \bar{z})$ encodes the non-trivial dynamics, namely the exchange of long multiplets, which are dual to stringy states in the bulk theory. From $H^{I_1 I_2 I_3 I_4}(z, \bar{z})$ we define the reduced Mellin amplitude

$$H^{I_1 I_2 I_3 I_4}(z, \bar{z}) = \int_{-i\infty}^{i\infty} \frac{ds dt}{(4\pi i)^2} |z|^{s+2} |1 - z|^{t-2} \times \Gamma(s, t) M^{I_1 I_2 I_3 I_4}(s, t),$$

$$\Gamma(s, t) = \Gamma\left[1 - \frac{s}{2}\right]^2 \Gamma\left[1 - \frac{t}{2}\right]^2 \Gamma\left[1 - \frac{u}{2}\right]^2,$$

with $s + t + u = 0$. In this letter, we consider open string scattering where the worldsheet has the topology of a disk, with four insertions at its boundary, in an expansion around flat space. This corresponds to the leading nontrivial order in a large central charge expansion of the Mellin amplitude,

* Contact author: luis.alday@maths.ox.ac.uk

† Contact author: rs.pitombo@unesp.br

‡ Contact author: aurelie.sangare@maths.ox.ac.uk

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while rescaling $M^{I_1 I_2 I_3 I_4}(s, t) \rightarrow \lambda^{3/2} M^{I_1 I_2 I_3 I_4}(\sqrt{\lambda}s, \sqrt{\lambda}t)$ and expanding in $1/\sqrt{\lambda}$ with the rescaled variables fixed. At this order the color structure of the correlator is given by the sum of single traces of products of generators T^I of G_F , and we can decompose $M^{I_1 I_2 I_3 I_4}(s, t)$ into three color-ordered amplitudes

$$\begin{aligned} M^{I_1 I_2 I_3 I_4}(s, t) &= \text{Tr}(T^{I_1} T^{I_2} T^{I_3} T^{I_4}) M(s, t) \\ &+ \text{Tr}(T^{I_1} T^{I_4} T^{I_2} T^{I_3}) M(t, u) \\ &+ \text{Tr}(T^{I_1} T^{I_3} T^{I_4} T^{I_2}) M(u, s). \end{aligned}$$

As shown, the three independent color structures are related by crossing. In addition $M(s, t) = M(t, s)$. The AdS Veneziano amplitude $A(S, T)$ is related to the color-ordered Mellin amplitude $M(s, t)$ by the Borel transform

$$M(s, t) = \frac{8}{\lambda} \int_0^\infty d\beta e^{-\beta} \beta^3 A\left(\frac{2\beta s}{\sqrt{\lambda}}, \frac{2\beta t}{\sqrt{\lambda}}\right). \quad (1)$$

From the perspective of scattering amplitudes on AdS it is natural to use the AdS radius R , related to the t' Hooft coupling λ at large central charge by $\sqrt{\lambda} = R^2/\alpha'$. The AdS amplitude admits an expansion around flat space

$$A(S, T) = A^{(0)}(S, T) + \sum_{k=1}^\infty \left(\frac{\alpha'}{R^2}\right)^k A^{(k)}(S, T). \quad (2)$$

Note that at zeroth order, we recover the usual formula for the flat-space limit of holographic correlators [8], so that it is natural to expect that (1) should reproduce the AdS Veneziano amplitude in a small curvature expansion. In particular, $A^{(0)}(S, T)$ is the Veneziano amplitude in flat space

$$A^{(0)}(S, T) = \frac{1}{U} \int_0^1 x^{S-1} (1-x)^{T-1} dx, \quad (3)$$

with $S + T + U = 0$. At higher orders in the curvature expansion the proposed structure is the following

$$A^{(k)}(S, T) = \frac{1}{U} \int_0^1 x^{S-1} (1-x)^{T-1} g^{(k)}(S, T; x) dx, \quad (4)$$

with $g^{(k)}(S, T; x)$ given by a linear combination of multiple polylogarithms (MPLs) $L_w(x)$, labeled by words in the alphabet $\{0, 1\}$ up to length $3k$. More precisely

$$g^{(k)}(S, T; x) = \frac{1}{(S+T)^k} \sum_{n=0}^{3k} \sum_{|w|=n} P_w^k(S, T) L_w(x), \quad (5)$$

with $P_w^k(S, T)$ homogeneous polynomials of degree $|w|$ (and labeled by both w and k). MPLs $L_w(x)$ are recursively defined as follows. For the empty word $L_e(x) = 1$. For a general word

$$\frac{d}{dx} L_{aw}(x) = \frac{1}{x-a} L_w(x), \quad a = 0, 1, \quad (6)$$

together with $L_w(0) = 0$ for $w \neq 0^p$ and $L_{0^p}(x) = \frac{1}{p!} \log^p x$. The length of the word, denoted by $|w|$, is called the weight [9]. Crossing symmetry implies

$$g^{(k)}(S, T; x) = g^{(k)}(T, S; 1-x). \quad (7)$$

The integrand is known for $k = 0, 1, 2$, see [6, 7].

Monodromy relations in AdS. String theory color-ordered disk amplitudes in flat space satisfy linear relations known as monodromy relations [10–12]. For the Veneziano amplitude in flat space

$$e^{\pm i\pi S} A^{(0)}(S, U) + A^{(0)}(S, T) + e^{\mp i\pi T} A^{(0)}(U, T) = 0, \quad (8)$$

where recall $S + T + U = 0$. In flat space, these relations can be derived directly from the worldsheet perspective. Indeed, consider the following integrals

$$\int_{\mathcal{C}^\pm} \eta(z) dz = 0, \quad \eta(z) = z^{S-1} (1-z)^{T-1} \quad (9)$$

where the contours $\mathcal{C}^\pm$ are given by straight lines just above/below the real line. The integrand $\eta(z)$ is a multi-valued function on the complex plane, with branch cuts along the real line for $x \leq 0$ and $x \geq 1$. We define it such that

$$\eta(x \pm i\epsilon) = \begin{cases} -e^{\pm i\pi S} (-x)^{S-1} (1-x)^{T-1} & \text{for } x < 0, \\ x^{S-1} (1-x)^{T-1} & \text{for } 0 < x < 1, \\ -e^{\mp i\pi T} (x)^{S-1} (x-1)^{T-1} & \text{for } x > 1. \end{cases} \quad (10)$$

When integrating over the regions $(-\infty, 0]$, $[0, 1]$ and $[1, \infty)$ we recover the three color-ordered amplitudes $A^{(0)}(S, U)$, $A^{(0)}(S, T)$, $A^{(0)}(U, T)$ respectively, and the monodromy relations. Furthermore, note that different color-ordered amplitudes correspond to different orderings of the vertex operators inserted at the boundary of the disk, so that for instance

$$A^{(0)}(S, T) \sim \int_0^1 dx \langle V_1(0) V_2(x) V_3(1) V_4(\infty) \rangle. \quad (11)$$

The phases present in the monodromy relations can be seen as the result of exchanging the order of the vertex operators. Let us now analyze the situation in AdS. The phase $e^{\pm i\pi S}$ is directly related to the monodromy of z^S as we move around the origin. The phase $e^{\mp i\pi T}$ is directly related to the monodromy of $(1-z)^T$ as we move around $z = 1$. The presence of MPLs in $g^{(k)}(S, T; x)$ gives extra contributions as we

move around zero and one. Let us denote by K_0 the operator such that the monodromy of MPLs around zero is given by the action of $e^{2\pi i K_0}$. Its action on MPLs is given by

$$K_0 L_{wa}(x) = L_w(x) \delta_{a0}, \quad K_0 L_e(x) = 0. \quad (12)$$

In the same way, we introduce K_1 so that the monodromy around one is given by the action of $e^{2\pi i K_1}$. Its precise action on MPLs involves the Drinfeld associator and will be given below. We are now ready to write down the expected monodromy relations in AdS. Taking into account the monodromies of the prefactor $\eta(z)$ as well as the monodromies of the extra insertion $g(S, T; x)$ we obtain

$$e^{\pm i\pi(S+K_0)} A(S, U) + A(S, T) + e^{\mp i\pi(T+K_1)} A(U, T) = 0. \quad (13)$$

Several comments are in order. First, these relations are a very clear manifestation of—and were possible to guess thanks to—the worldsheet representation for the open string amplitude on AdS. Indeed, the action of K_0, K_1 is only defined at the level of the worldsheet integrand: they act nontrivially on MPLs, but trivially on rational functions of S and T . Second, these proposed relations should hold order by order in the curvature expansion, and they do not mix different orders. Lastly, the action of K_0, K_1 on the flat-space result is trivial, and hence in this case we obtain the usual monodromy relations. We can also use crossing symmetry to write down the monodromy relations purely in terms of K_0

$$e^{\pm i\pi(S+K_0)} A(S, U) + A(S, T) + e^{\mp i\pi(T+K_0)} A(T, U) = 0. \quad (14)$$

Given the structure of the integrands, we can introduce the building blocks of the amplitude [13], a family of linearly independent integrals labeled by words w

$$J_w(S, T) = \int_0^1 x^{S-1} (1-x)^{T-1} L_w(x) dx, \quad (15)$$

on which the operator K_0 acts as

$$K_0 J_{wa}(S, T) = J_w(S, T) \delta_{a0}, \quad K_0 J_e(S, T) = 0, \quad (16)$$

while leaving rational functions of S and T invariant. In order to understand the consequences of the monodromy relations (14), we need to work out the properties of the building blocks $J_w(S, T)$ under crossing of the AdS Mandelstam variables. It is convenient to introduce non-commuting variables (e_0, e_1) and a generating function for MPLs

$$L(e_0, e_1; x) = \sum_w L_w(x) e_w, \quad (17)$$

with $e_w = 1$ for the empty word and for instance $e_{0011} = e_0 e_0 e_1 e_1$. This induces a generating function for the J integrals

$$\mathcal{J}(S, T; e_0, e_1) = \int_0^1 x^{S-1} (1-x)^{T-1} L(e_0, e_1; x) dx. \quad (18)$$

Inspired by the monodromy relations in flat space, we will work out linear relations between $\mathcal{J}(S, T; e_0, e_1)$ and its crossing cousins. We start by considering the following integrals

$$\int_{\mathcal{C}^\pm} \eta(z) L(e_0, e_1; z) dz = 0, \quad (19)$$

where for $\mathcal{C}^\pm$ we have $z = x \pm i\epsilon$, just above/below the real line. MPLs have branch cuts along the real line for $x < 0$ and $x > 1$, so we have to be careful when doing contour manipulations. Splitting the contour of integration into three segments and making a simple change of variables we obtain

$$\begin{aligned} \mathcal{J}(S, T; e_0, e_1) & - e^{\pm i\pi S} \int_0^1 x^{-S-T} (1-x)^{S-1} L\left(e_0, e_1; 1 - \frac{1}{x} \pm i\epsilon\right) dx \\ & - e^{\mp i\pi T} \int_0^1 x^{-S-T} (1-x)^{T-1} L\left(e_0, e_1; \frac{1}{x} \pm i\epsilon\right) dx = 0. \end{aligned}$$

We have used (10) and changed variables to bring the regions of integration to $x \in [0, 1]$ in all cases. Note that for $x \in [0, 1]$ the generating functions in the second and third lines are evaluated just above/below the branch cuts. They can be related back to the generating function evaluated at x by the identities [14]

$$\begin{aligned} L\left(e_0, e_1; 1 - \frac{1}{x} \pm i\epsilon\right) & = L(-e_0 - e_1, e_0; x) \\ & \quad \times Z(e_0, -e_0 - e_1) e^{\pm i\pi e_0}, \\ L\left(e_0, e_1; \frac{1}{x} \pm i\epsilon\right) & = L(-e_0 - e_1, e_1; x) \\ & \quad \times Z(e_1, -e_0 - e_1) e^{\mp i\pi e_1} Z(e_0, e_1), \end{aligned}$$

where the Drinfeld associator $Z(e_0, e_1)$ can be defined as the regularized generating function at $x = 1$

$$Z(e_0, e_1) \equiv L(e_0, e_1; 1) = 1 - \zeta(2)[e_0, e_1] + \dots \quad (20)$$

and satisfies $Z(e_0, e_1) Z(e_1, e_0) = 1$. This leads to

$$\begin{aligned} \mathcal{J}(S, T; e_0, e_1) - \mathcal{J}(S, 1 - S - T; e_0, -e_0 - e_1) e^{\pm i\pi(S+e_0)} \\ - \mathcal{J}(T, 1 - S - T; e_1, -e_0 - e_1) e^{\mp i\pi(T+e_1)} Z(e_0, e_1) = 0, \end{aligned} \quad (21)$$

where we have used

$$\mathcal{J}(S, T; e_1, e_0) Z(e_0, e_1) = \mathcal{J}(T, S; e_0, e_1).$$

Now combine (21) with the shift relations found in [13]

$$\mathcal{J}(S + 1, T; e_0, e_1) = U_s \mathcal{J}(S, T; e_0, e_1), \quad (22)$$

$$\mathcal{J}(S, T + 1; e_0, e_1) = U_t \mathcal{J}(S, T; e_0, e_1) \quad (23)$$

with $U_s = \frac{1}{S+T+e_0+e_1}(S+e_0)$, $U_t = 1 - U_s$. This leads to

$$\begin{aligned} \mathcal{J}(S, T; e_0, e_1) - U_t^{-1} \mathcal{J}(S, U; e_0, -e_0 - e_1) e^{\pm i\pi(S+e_0)} \\ - U_s^{-1} \mathcal{J}(T, U; e_1, -e_0 - e_1) e^{\mp i\pi(T+e_1)} Z(e_0, e_1) = 0. \end{aligned} \quad (24)$$

With this at hand, we can plug a given proposal for the amplitude into (14), and then use (24) to express the monodromy relations in terms of a set of conditions, each multiplying an independent building block $J_w(S, T)$. As we will see below, this gives highly nontrivial constraints on the amplitude.

At the level of generating functions, K_0 and K_1 act as

$$\begin{aligned} K_0 L(e_0, e_1; x) &= L(e_0, e_1; x) e_0, \\ K_1 L(e_0, e_1; x) &= L(e_0, e_1; x) Z(e_1, e_0) e_1 Z(e_0, e_1) \end{aligned}$$

and similarly when acting on the generating function $\mathcal{J}(S, T; e_0, e_1)$. We then see that the monodromies around $x = 0$ and $x = 1$, computed in [15], are given by

$$\begin{aligned} L(e_0, e_1; x + i\epsilon) &= e^{2\pi i K_0} L(e_0, e_1; x - i\epsilon), & x < 0, \\ L(e_0, e_1; x - i\epsilon) &= e^{2\pi i K_1} L(e_0, e_1; x + i\epsilon), & x > 1. \end{aligned}$$

Implications. Let us consider the first curvature correction to the Veneziano amplitude

$$A^{(1)}(S, T) = \sum_w \frac{P_w(S, T)}{U^2} J_w(S, T), \quad (25)$$

where the sum runs over words of weight up to three, and $P_w(S, T)$ are homogeneous polynomials of degree $|w|$. The general ansatz consistent with crossing symmetry has 33 unfixed parameters. The monodromy relations give 28 independent linear constraints, leading to only five independent solutions. As expected, the amplitude found in [6] is among them. There is no solution with

transcendentality one, see Supplemental Material [16]. The simplest solution has transcendentality two,

$$\begin{aligned} A^{(1)}(S, T) &= \frac{J_e(S, T) - T J_0(S, T) - S J_1(S, T)}{U^2} \\ &+ \frac{S}{U} J_{00}(S, T) + \frac{T}{U} J_{11}(S, T) + J_{01}(S, T) \\ &+ J_{10}(S, T). \end{aligned}$$

Finally, we also note that all five solutions satisfy the following constraints:

$$\begin{aligned} P_{01}(S, T) &= P_{10}(S, T), \\ P_{110}(S, T) &= P_{101}(S, T), \quad P_{001}(S, T) = P_{010}(S, T). \end{aligned}$$

Namely, the MPLs that arise are the diagonal limit of single-valued MPLs. For more complicated examples we need to develop an efficient way to solve the monodromy relations. Introduce a linear operator $\Psi: e_w \rightarrow \mathbb{R}$ such that

$$A(S, T) = \langle \Psi(S, T; f_0, f_1) | \mathcal{J}(S, T; e_0, e_1) \rangle. \quad (26)$$

This is achieved by defining

$$\Psi(S, T; f_0, f_1) = \sum_w R_{\tilde{w}}(S, T) f_w, \quad (27)$$

with the notation $f_e = 1, f_{001} = f_0 f_0 f_1$ and so on, together with an inner product in the space of words

$$\langle f_{\tilde{w}} | e_{w'} \rangle = \delta_{ww'}, \quad (28)$$

where $\tilde{w}$ is the operation that reverses the order of the letters. In this language we can write the monodromy relations as

$$\begin{aligned} &\langle \Psi(S, T; f_0, f_1) | \mathcal{J}(S, T; e_0, e_1) \rangle \\ &+ \langle \Psi(S, U; f_0, f_1) | \mathcal{J}(S, U; e_0, e_1) e^{\pm i\pi(S+e_0)} \rangle \\ &+ \langle \Psi(U, T; f_0, f_1) | \mathcal{J}(T, U; e_1, e_0) e^{\mp i\pi(T+e_1)} Z(e_0, e_1) \rangle = 0. \end{aligned} \quad (29)$$

The difference between the two linear relations (24) gives

$$\begin{aligned} \mathcal{J}(S, U; e_0, e_1) &= K \mathcal{J}(T, U; -e_0 - e_1, e_1) \\ &\times \sin \pi(T - e_0 - e_1) Z(e_0, -e_0 - e_1) \\ &\times (\sin \pi(S + e_0))^{-1}, \end{aligned} \quad (30)$$

with

$$K = \frac{1}{S + T - e_1} (T - e_0 - e_1) \frac{1}{S + e_0} (S + T - e_1).$$

Plugging (30) into the difference of the two monodromy relations (29) we obtain

$$\begin{aligned} \langle \Psi(S, U; f_0, f_1) | K \tilde{\mathcal{J}}(T, U; -e_0 - e_1, e_1) Z(e_0, -e_0 - e_1) \rangle \\ = \langle \Psi(U, T; f_0, f_1) | \tilde{\mathcal{J}}(T, U; e_1, e_0) Z(e_0, e_1) \rangle \end{aligned}$$

where $\tilde{\mathcal{J}}(T, U; e_1, e_0) = \mathcal{J}(T, U; e_1, e_0) \sin \pi(T + e_1)$. Using the properties for the inner product given in Sec. II of Supplemental Material [16] this turns into

$$\begin{aligned} \langle \Psi(S, U; -f_0, -f_0 + f_1) | \tilde{\mathcal{K}} \tilde{\mathcal{J}}(T, U; e_0, e_1) Z(-e_0 - e_1, e_0) \rangle \\ = \langle \Psi(U, T; f_1, f_0) | \tilde{\mathcal{J}}(T, U; e_0, e_1) Z(e_1, e_0) \rangle \end{aligned}$$

with

$$\tilde{\mathcal{K}} = \frac{1}{S + T - e_1} (T + e_0) \frac{1}{S - e_0 - e_1} (S + T - e_1).$$

Since these equations are valid for any basis of independent functions $\tilde{\mathcal{J}}(T, U; e_0, e_1)$ we must have

$$\begin{aligned} Z(-\tau_0 - \tau_1, \tau_0) \Psi(S, U; -f_0, -f_0 + f_1) \tilde{\mathcal{K}} \\ = Z(\tau_1, \tau_0) \Psi(U, T; f_1, f_0), \end{aligned} \quad (31)$$

where the operator $\tilde{\mathcal{K}}$ acts from the right [17]

$$\tilde{\mathcal{K}} = \frac{1}{S + T - \tau_1} (T + \tau_0) \frac{1}{S - \tau_0 - \tau_1} (S + T - \tau_1). \quad (32)$$

Combining crossing symmetry $A(S, T) = A(T, S)$ with $\mathcal{J}(T, S; e_0, e_1) = \mathcal{J}(S, T; e_1, e_0) Z(e_0, e_1)$ we obtain

$$\Psi(S, T; f_0, f_1) = Z(\tau_1, \tau_0) \Psi(T, S; f_1, f_0), \quad (33)$$

which combined with (31) leads to

$$\begin{aligned} Z(-\tau_0 - \tau_1, \tau_0) \Psi(S, U; -f_0, -f_0 + f_1) \tilde{\mathcal{K}} \\ = \Psi(T, U; f_0, f_1). \end{aligned} \quad (34)$$

Equations (33) and (34) do not make any reference to the integrals $J_w(S, T)$, but they are equivalent, of course, to crossing symmetry and the monodromy relations presented before. One can explicitly check that the first two curvature corrections found in [6,7], $\Psi^{(1)}(S, T; f_0, f_1)$ and $\Psi^{(2)}(S, T; f_0, f_1)$, satisfy these equations. At a given order in the small curvature expansion we have

$$\Psi^{(k)}(S, T; f_0, f_1) = \sum_{t=0}^{3k} \Psi_t^{(k)}(S, T; f_0, f_1) \quad (35)$$

where $\Psi_t^{(k)}(S, T; f_0, f_1)$ denotes the weight t contribution. The maximally transcendental piece with $t = 3k$ satisfies a

simplified system of equations

$$\begin{aligned} \Psi_{3k}^{(k)}(S, T; f_0, f_1) &= Z(\tau_1, \tau_0) \Psi_{3k}^{(k)}(T, S; f_1, f_0), \\ &\times \frac{\Psi_{3k}^{(k)}(S, U; -f_0, -f_0 + f_1)}{S} \\ &= Z(\tau_0, -\tau_0 - \tau_1) \frac{\Psi_{3k}^{(k)}(T, U; f_0, f_1)}{T}, \end{aligned} \quad (36)$$

since any higher orders in the operator $\tilde{\mathcal{K}}$ would lower the weight. Consistency with the high-energy limit of the amplitudes $A^{(k)}(S, T)$ —large S, T with S/T fixed—imposes

$$\Psi_{3k}^{(k)}(S, T; f_0, f_1) = \frac{U^{k-1}}{k!} \left(\Psi_3^{(1)}(S, T; f_0, f_1) \right)^{\sqcup, k} \quad (37)$$

where the notation $(\cdot)^{\sqcup, k}$ means shuffle product with itself, k times. As a result of the relations (S10) in Supplemental Material [16], one can easily show that (37) satisfies both equations in (36) for general k , provided the result holds for $k = 1$. Hence the monodromy relations are consistent with the high-energy limit, to all orders in the curvature expansion.

Monodromy relations for general KK modes. The first curvature correction to the AdS Veneziano amplitude corresponding to the scattering of arbitrary Kaluza-Klein (KK) modes on the sphere was constructed in [18]. The amplitude is labeled by the KK modes p_1, p_2, p_3, p_4 and reduces to the one considered in [6,7] for $p_i = 2$. The amplitude depends on the usual AdS Mandelstam variables S, T, U , with $S + T + U = 0$, and on the sphere Mandelstam variables N_s, N_t, N_u , with $N_s + N_t + N_u = -1$. For general KK modes we expect the monodromy relations to take the form

$$\begin{aligned} e^{\pm i\pi(S+K_0)} A_{KK}(S, U, N_s, N_u) + A_{KK}(S, T, N_s, N_t) \\ + e^{\mp i\pi(T+K_0)} A_{KK}(T, U, N_t, N_u) = 0. \end{aligned} \quad (38)$$

The full expression taken from [18] can be found in Sec. III of Supplemental Material [16]. One can explicitly check that it satisfies (38).

Conclusions. Over the last few years methods have been developed to compute tree-level string theory amplitudes on AdS spaces [19,20]. One of the most striking features of these results is the emergence of a worldsheet picture [21,22], both for closed as well as for open string amplitudes. In this letter we propose a set of monodromy relations for open string amplitudes in AdS. These arise as a consequence of the worldsheet description, are satisfied order by order in the small curvature expansion, and in flat space they reduce to the usual monodromy relations for the

Veneziano amplitude. The monodromy relations are very powerful in flat space, see [23–25], and the same appears to be true in AdS. In the context of the proposal of [6,7], and when combined with crossing symmetry alone, they reduce the number of free parameters from 33 to 5 for the first curvature correction, and from 565 to 86 for the second curvature correction. A natural direction is to use these new constraints to study and compute further curvature corrections. It would be interesting to understand the monodromy relations found in this letter in terms of braiding relations for open string vertex operators in AdS backgrounds, along the lines of [26,27]. More generally, it

would be very interesting to understand how the structure studied in this letter arises from a first-principles world-sheet computation.

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 - [16] See Supplemental Material at <http://link.aps.org/supplemental/10.1103/mppb-xh1j> for a toy model illustrating the constraints imposed by the monodromy relations (Sec. I); properties of the monodromy linear operator (Sec. II); and the explicit expression for the first curvature correction to the AdS Veneziano amplitude for arbitrary KK modes (Sec. III).
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$$\tau_i f_{jw} = \delta_{ij} f_w, \quad f_{wj} \tau_i = \delta_{ij} f_w.$$
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